

KABISH SRIDAR

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EDUCATION

SRM Institute of Science and Technology (SRMIST) | Tiruchirappalli, India

Bachelor of Technology in Computer Science and Engineering with specialization in Artificial Intelligence and Machine Learning

(Expected Graduation: May 2029)

- **Current CGPA: 8.45**

TECHNICAL SKILLS

- **Languages:** Python, C, C++, SQL (PostgreSQL, MySQL, SQLite), Structured Text (IEC 61131-3)
- **Frameworks & Libraries:** Flask, OpenCV, DeepFace, YOLOv8, PyTorch
- **DevOps & Tools:** Docker, Docker Compose, Git, Ubuntu, Redis (Caching)
- **Automation Hardware:** Raspberry Pi 4, Arduino, ABB AC500 PLC (Automation Builder)

TECHNICAL PROJECTS

Gap Measurement Vision Pipeline

- **Engineering & Algorithm:** Developed an automated image processing pipeline using OpenCV contouring and perspective transforms to calculate physical industrial edge tolerances dynamically from live video feeds at low cost with a Raspberry Pi, a PiCamera, and OpenCV image processing.
- **Problem Solved & Outcome:** Replaced subjective, slow manual caliper checks with a hands-free, automated system that yields sub-millimeter measurement consistency, drastically accelerating QA inspection speeds.

Object Detection Deployment via Containerized Pipeline

- **Engineering & DevOps:** Built an end-to-end YOLOv8 real-time object detection inference pipeline containerized via multi-stage Dockerfiles and orchestrated using Docker Compose to bridge host camera hardware.
- **Problem Solved & Outcome:** Eliminated environment variance across different host OS platforms, providing development teams with a 100% reproducible deployment setup that cuts edge-device testing cycles from hours to minutes.

EMO-REX – Emotion Recognition System

- **Engineering & Data Layer:** Engineered a high-concurrency facial expression analysis app using OpenCV and DeepFace, backed by an in-memory Redis cache for fast state lookups and PostgreSQL for persistent analytics.
- **Problem Solved & Outcome:** Resolved severe database write bottlenecks during continuous live feeds by reducing state-lookup latency by 40%, ensuring fluid, crash-free user tracking and reliable metric logging.

Automated KYC Verification Platform

- **Engineering & Data Layer:** Architected a secure Know Your Customer onboarding engine leveraging a dual-database pattern (SQLite for fast session persistence and PostgreSQL for master data storage).
- **Problem Solved & Outcome:** Solved user drop-off bottlenecks by optimizing concurrent transaction write cycles, delivering instant validation processing and an airtight data pipeline that eliminates timeout failures.

Industrial Oil Batching Automation System

- **Engineering & Automation:** Programmed a fluid scaling system using an ABB AC500 PLC, authoring low-latency Structured Text (ST) logic in ABB Automation Builder V2.0 to monitor digital/analog I/O feedback loops.
- **Problem Solved & Outcome:** Removed manual override dependencies by 95% in the mixing phase, eliminating human error, preventing equipment overflows, and guaranteeing absolute batch consistency.